



***ERBB3* Overexpression is Enriched in Diverse Patient Populations with Castration-sensitive Prostate Cancer and is Associated with a Unique AR Activity Signature**

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ABSTRACT

Purpose: Despite successful clinical management of castration-sensitive prostate cancer (CSPC), the 5-year survival rate for men with castration-resistant prostate cancer is only 32%. Combination treatment strategies to prevent disease recurrence are increasing, albeit in biomarker-unselected patients. Identifying a biomarker in CSPC to stratify patients who will progress on standard-of-care therapy could guide therapeutic strategies.

Experimental Design: Targeted deep sequencing was performed for the University of Illinois (UI) cohort ($n = 30$), and immunostaining was performed on a patient tissue microarray ($n = 149$). Bioinformatic analyses identified pathways associated with biomarker overexpression (OE) in the UI cohort, consolidated RNA sequencing samples accessed from Database of Genotypes and Phenotypes ($n = 664$), and GSE209954 ($n = 68$). Neutralizing antibody patritumab and ectopic HER3 OE were utilized for functional mechanistic experiments.

Results: We identified *ERBB3* OE in diverse patient populations with CSPC, where it was associated with advanced disease

at diagnosis. Bioinformatic analyses showed a positive correlation between *ERBB3* expression and the androgen response pathway despite low dihydrotestosterone and stable expression of androgen receptor (AR) transcript in Black/African American men. At the protein level, HER3 expression was negatively correlated with intraprostatic androgen in Black/African American men. Mechanistically, HER3 promoted enzalutamide resistance in prostate cancer cell line models and HER3-targeted therapy resensitized therapy-resistant prostate cancer cell lines to enzalutamide.

Conclusions: In diverse patient populations with CSPC, *ERBB3* OE was associated with high AR signaling despite low intraprostatic androgen. Mechanistic studies demonstrated a direct link between HER3 and enzalutamide resistance. *ERBB3* OE as a biomarker could thus stratify patients for intensification of therapy in castration-sensitive disease, including targeting HER3 directly to improve sensitivity to AR-targeted therapies.

Introduction

Prostate cancer is commonly diagnosed among men in the United States, resulting in >34,000 deaths per year (1). The 5-year survival rate for localized disease remains high; however, mortality increases substantially in metastatic disease, with a 5-year survival of only 32% (1). As an androgen-driven disease, prostate cancer is clinically managed with therapies targeting the androgen receptor (AR), including androgen deprivation therapy and androgen receptor signaling inhibitors (ARSI). While these therapies are initially successful, many cases progress to

castration-resistant prostate cancer (CRPC). There are several established mechanisms of progression to CRPC, with most cases occurring when AR gains functionality through mutation, increased copy number, ligand promiscuity, the emergence of ligand-independent variants, or retention of AR signaling through alternative receptors/co-factors (2–4). Verified risk factors for advancement to CRPC include age, family history, and race (5). This is clinically significant given the racial disparity in prostate cancer mortality observed in Black/African American (AA) men, where mortality rates are two to four times higher than those in every other racial and ethnic group in the United States (1, 6). Unfortunately, despite our improved understanding of molecular changes and risk factors associated with CRPC, survival rates remain low (3). While combination treatment strategies to prevent disease recurrence are moving into the frontline for the management of advanced disease, recent landmark prospective studies have been performed in biomarker-unselected patients (7, 8). This unselected intensification strategy may ultimately result in overtreatment in a subset of patients. Therefore, identifying a predictive biomarker in CSPC for the subset of patients that will progress on antiandrogen therapy could guide intelligent intensification to optimize targeted therapeutic strategies in those most vulnerable to developing aggressive, resistant disease.

HER3 (*ERBB3*) has recently emerged as a potential actionable target for advanced prostate cancer, where it was correlated to shorter time to castration resistance and shorter overall survival (9). In preclinical studies, direct HER3 inhibition or indirect targeting through its ligand NRG1 decreased proliferation and migration *in vitro* and decreased tumor volume *in vivo* (9–11). Notably, the scope of these studies is limited to targeting HER3 in advanced CRPC. However, HER3

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Translational Relevance

ERBB3 (HER3) in prostate cancer is associated with poor clinical outcomes, including shorter time to castration resistance and reduced overall survival. Prior studies have proposed its role in androgen receptor signaling and postulated it as an actionable target in castration-resistant prostate cancer. In this study, *ERBB3* overexpression was found to be prevalent in racially diverse men with castration-sensitive disease. *ERBB3* mRNA expression was associated with increased androgen receptor signaling and high serum PSA, and at the protein level, HER3 was associated with low intraprostatic androgen, specifically in Black/African American men. Further mechanistic studies demonstrate a direct relationship between HER3 expression and enzalutamide sensitivity. We posit that *ERBB3* overexpression may have prognostic significance in assisting clinicians with therapy selection, particularly in the castration-sensitive setting. Including *ERBB3* as a biomarker in prospective trials could help stratify aggressive disease and guide intelligent intensification of therapy.

has also been implicated in castration-sensitive prostate cancer (CSPC), where several studies suggest cross-talk between HER3 and AR signaling (12–14). Because AR-targeted therapies are standard-of-care for clinical management of CSPC, signaling mechanisms between HER3 and AR could be highly relevant to disease progression and the development of therapeutic resistance; the clinical implications of this cross-talk, however, have not been fully established. In this study, we identified racially diverse patient populations with CSPC with prevalent *ERBB3* overexpression (OE); in the University of Illinois (UI) cohort and others, *ERBB3* RNA OE was enriched in Black/AA men and was associated with aggressive disease at diagnosis. Considering the enrichment of *ERBB3* OE in Black/AA patients and the established cross-talk between *ERBB3* and AR, we hypothesized that *ERBB3* OE could provide prognostic value as a biomarker for intensification of therapy in Black/AA men with CSPC and that this pathway can be targeted to improve prostate cancer outcomes.

Materials and Methods

Sample collection and sequencing

Treatment-naïve tumor samples were collected from patients in the UI cohort through biopsy or radical prostatectomy at UI Health (Table 1). Targeted deep sequencing was performed on banked tissue samples as part of routine standard of care through Tempus Laboratories (15, 16). The decision to send samples for sequencing was based on the following criteria: (i) patients received care at UI Health for advanced prostate cancer between January 2022 and July 2022, (ii) patients were eligible for sequencing per National Comprehensive Cancer Network (NCCN) Guidelines on Prostate Cancer V.1.2022, and (iii) patients had tissue banked that was available to send. In total, 65 patient samples were sent for sequencing during this period; however, only 33 patients had sufficient tissue quantity and quality for RNA sequencing (RNA-seq). Deidentified raw FASTQ sequencing data were obtained from Tempus and stored in a secure server with password encryption. Retrospective data analysis was performed on deidentified data after approval by an Institutional Review Board (IRB; IRB-STUDY2022-0907) and in accordance with the U.S. Department of Health and Human

Table 1. Patient characteristics in the UI Health cohort and dbGaP samples.

	UI cohort <i>n</i> (%)	dbGaP samples <i>n</i> (% of reported)
Sample size	33	541
Age at baseline (years)		
<i>Percent reported</i>	100%	15.0%
Median	63.4	72
Range	51.9–77.7	39–92
Self-reported race		
<i>Percent reported</i>	100%	14.8%
Black/AA	23 (69.7)	1 (1.25)
White	7 (21.2)	77 (96.3)
Unknown	2 (6.0)	0 (0)
American Indian	1 (3.0)	1 (1.25)
Asian	0 (0)	1 (1.25)
Self-reported ethnicity		
<i>Percent reported</i>	100%	0%
Non-Hispanic	27 (81.8)	
Hispanic	6 (18.2)	
Baseline PSA		
<i>Percent reported</i>	100%	4.62%
Median	226	5.3
Range	7.97–5800	2.8–13.7
Stage at diagnosis		
<i>Percent reported</i>	100%	0%
Localized (N0)	4 (12.1)	
Regional (N1)	6 (18.2)	
Metastatic (de novo)	23 (69.7)	
Tissue sequenced		
<i>Percent reported</i>	100%	100%
Localized tumor	33 (100)	101 (19.3)
Metastases		310 (59.2)
CRPC		115 (21.9)
NEPC		15 (2.86)
Gleason sum		
<i>Percent reported</i>	100%	4.62%
≤7	5 (15.2)	19 (76)
≥8	24 (72.3)	6 (24)
Unknown	4 (12.1)	0 (0)

Abbreviation: NEPC, neuroendocrine prostate cancer.

Services and the Declaration of Helsinki. A waiver was granted for written informed consent due to the retrospective nature of the study. Deidentification of data was maintained through all steps and generic identifiers were used to notate and analyze data.

Sequencing analysis and normalization

RNA FASTQ files were obtained from Tempus, and quality control of raw FASTQ data was determined using FastQC (<https://www.bioinformatics.babraham.ac.uk/projects/fastqc/>). Sequences were trimmed using cutadapt to remove TruSeq adapters and to quality trim to a minimum Phred score of 20 (17). Genome alignment was performed using STAR (RRID:SCR_004463) with the hg38 human genome as reference (18). Genomic features were quantified using FeatureCounts based on reverse-stranded libraries using default parameters against Ensembl annotations for human genome hg38 (19, 20). Normalized gene expression levels were computed using edgeR in counts per million (CPM) units, including the TMM (trimmed mean of M-values) normalization factor (21). Pairwise statistics between genes of interest using TPM (CPM-TMM) values was conducted using a Mann–Whitney U test.

Gene set enrichment analysis and AR signature score

TPM normalized RNA-seq data were sorted on *ERBB3* levels using inclusive quartiles where the *ERBB3*-high group are the patient IDs greater than or equal to the third quartile value and the *ERBB3*-low group are the patient IDs lower than or equal to the first quartile value. All genes associated with the high and low groups were extracted and gene set enrichment analysis (GSEA) comparing *ERBB3*-high and *ERBB3*-low was done using GSEA (SeqGSEA, RRID:SCR_005724) 4.3.2, as described previously (22). Settings altered from default in the program include number of permutations set to 1,000, permutation type set to phenotype, expression dataset to h.all.symbols.gmt[Hallmarks], Chip platform Human set to ENSEMBL_Gene_ID_MSigD8v7.5.1.chip, enrichment statistic set to weighted, and metric for ranking genes set to Signal2-Noise (23). Using the normalized RNA-seq data, an AR signature score was developed using a comprehensive list of AR target genes (24). The z-score $[(x - \mu)/\sigma]$ was calculated for each gene independently and the z-score for each AR target gene was summed to create a single AR signature score value per sample. This analysis was applied to all patients in the UI cohort.

Acquisition and analyses of publicly available RNA-seq datasets

Access to publicly available prostate cancer RNA-seq datasets was requested and approved via NIH Database of Genotypes and Phenotypes (dbGaP; refs. 25–30). IRB approval was obtained when necessary. FASTQ files were trimmed and aligned to the genome using the same method as described above. A total of 664 RNA-seq datasets were further analyzed. Genomic features were quantified using FeatureCounts as described above. Pairwise statistics between genes of interest using TPM (CPM-TMM) values was conducted using a Mann–Whitney U test.

The computational analysis for ancestry was performed as a *de novo* identification of admixture groups within the experimental population, followed by a mapping of those groups to predefined ancestries based on the GnoMAD database. All samples from publicly available RNA-seq samples and UI cohort were analyzed together. Variant calls for all RNA-seq samples were called against reference hg38 using bcftools, mpileup, and bcftools call (multiallelic caller, flag -m), resulting in 15,7109,372 total variants (31). Variant calls with <8x coverage were removed. Variants were filtered on the basis of the following criteria: (i) minor allele frequency >1%, (ii) sample call rate >30%, (iii) variant annotated in dbSNP, (iv) GnoMAD allele frequency >1% in at least one population, (v) not annotated in COSMIC or NCI60 (somatic mutations in cancer), and (vi) not annotated as an RNA editing site based on previously characterized RNA editing sites from REDportal (32). This resulted in 72,101 final variants. *De novo* admixture groups were obtained using Admixture, varying the number of admixture groups from $k = 2$ to 20, with the optimal number ($k = 5$) determined on the basis of the minimum error (33). Allele frequencies were compared with those from different ancestry groups in GnoMAD using Pearson correlations, and the closest match was putatively assigned as the ancestry label for each admixture group. Using the five-module SNP construct for ancestry, we categorized each patient into White [>65% non-Finnish European (NFE)], Black [>65% African (AFR)], or mixed (<65% of NFE and AFR). The mixed ancestry patients were excluded from further analysis, leaving a total of 524 samples. This ancestry model was validated in the UI Health Cohort, where patients that self-reported as Black/AA or White/Caucasian matched 88% with ancestry determination at ≥70% threshold and 100% with ancestry determination at >65% threshold (Supplementary Table S1). The

number of AFR patients identified from each dataset is shown in Supplementary Table S2.

Acquisition and analysis of GSE209954

Dataset GSE209954 was accessed using NCBI’s Gene Expression Omnibus (RRID:SCR_005012) interface (6). Gene expression values for *ERBB3* (identifier 3417249) were isolated from the series matrix file and stratified by race (African American—AAM, non-African American—NAAM). *ERBB3* expression was normalized using SCAN.UPC package from Bioconductor (RRID:SCR_006442), as described previously (6). Gleason score 4+3 samples were included for analysis. A simple linear regression comparing pretreatment PSA (ng/mL) with normalized *ERBB3* expression was calculated in GraphPad/Prism (v.9.4.1).

Tissue microarray

The tissue microarray (TMA) contains 1 mm cores from 149 UI Health radical prostatectomy patients (98 Black/AA, 51 White). A board-certified pathologist reviewed each core to confirm pathology and Gleason grade. For each patient, an adjacent area of the prostate was snap frozen and used to quantify intraprostatic testosterone (T) and dihydrotestosterone (DHT) levels by LC/MS-MS analyses, as described previously (34). Clinical data include tumor Gleason score, body mass index (BMI), and smoking status for each patient represented on the TMA (Table 2). This TMA has power >0.8 for an effect size of 0.13 by linear regression analyses.

Table 2. Patient characteristics in the TMA.

Categorical variables	n (%)	HER3 (log) Mean (SD)	P value
Race			
Black	98 (65.8)	2.74 (0.86)	0.55 ^a
White	51 (34.2)	2.81 (0.81)	
BMI			
Underweight	2 (1.3)	3.07 (0.21)	0.0050 ^b
Normal	35 (24.2)	2.90 (0.78)	
Overweight	53 (35.6)	2.85 (0.90)	
Obesity	58 (38.9)	2.60 (0.82)	
Gleason sum			
5	1 (0.7)	2.27 (0.78)	0.49 ^b
6	12 (8.1)	2.80 (0.83)	
7	116 (77.9)	2.73 (0.86)	
8	7 (4.7)	2.77 (0.84)	
9	13 (8.7)	3.03 (0.70)	
Smoking status			
Former	50 (33.6)	2.69 (0.80)	0.0048 ^c
Current	51 (34.2)	3.00 (0.79)	
Non-smoker	48 (32.2)	2.61 (0.88)	
Continuous variables	Mean (SD)	Correlation w/ HER3 (log) ^d	P value
log HER3	2.77 (0.84)		
Age	61 (6.5)	0.039	0.53
PSA	15.0 (44.0)	−0.066	0.29
log T (pg/mg)	1.93 (6.69)	0.163	0.0118
log DHT (pg/mg)	1.62 (1.31)	−0.066	0.31

^aTwo-sample *t* tests.

^bSpearman correlation.

^cANOVA *F*-tests.

^dPearson correlation coefficient.

IHC staining and analysis

A total of 5-µm sections of the prostate cancer TMA were deparaffinized and stained on BOND RX automated stainer according to the BOND Polymer Refine Detection Kit (Leica Biosystems, #DS9800) protocol. Briefly, after deparaffinization, sections were subjected to heat-based antigen retrieval with BOND Epitope Retrieval Solution 2 (pH 9.0, Leica Biosystems, #AR9640) for 20 minutes at 100°C. Endogenous peroxidase activity and non-specific binding was blocked by sequentially treating samples with peroxidase block (BOND Polymer Refine Detection Kit) and Background sniper protein block (Biocare Medical, #BS966) for 15 minutes at room temperature. Sections were then incubated with anti-HER3/ERBB3 antibody (1:100, Cell Signaling Technology #12708, RRID:AB_2721919) for 60 minutes, and signal was detected using DAB (3,3'-Diaminobenzidine). All sections were counterstained with hematoxylin and mounted with Micromount media (Leica Microsystems, #3801730).

Slides were scanned at 20x resolution on an AT2 scanner and image analysis was performed using HALO version 3.5 and HALO AI (Indica Labs). Whole-slide images were segmented into individual TMA cores using the HALO TMA module. A pathologist reviewed all cores and identified regions of each core that were not consistent with the tissue description in the TMA key. Those regions were excluded from analysis. Within each core, tissue areas were identified, and artifacts were excluded using a modified version of the HALO AI QC Slide V2 classifier. Epithelial regions were identified via a custom-trained HALO AI MiniNet classifier. The HALO Area Quantification Module was used to separate DAB stain for ERBB3 using color deconvolution and set a threshold for positive staining. In epithelial-classified areas, each pixel was compared with the ERBB3 staining threshold and percentage of the area positive for the stain was reported.

Statistical analysis of TMA

Statistical analysis of the TMA was done in collaboration with the UI Cancer Center Biostatistics Shared Resource Core. Because of skewness of the HER3 and hormone distributions, log-transformation was applied when necessary. *t* tests or ANOVA models were used to compare HER3 expressions between demographic categories. Spearman or Pearson correlations between log-transformed HER3 and ordinal (BMI, Gleason sum) or continuous variables (age, PSA, androgen levels T and DHT). Multivariate regression models with backward model selection method were employed to identify factors associated with (log) HER3. Interactions between androgens and race were tested in the identifications of factors that modified the correlations between androgens and HER3. All statistical tests were two sided, controlling for type I error probability of 0.05. Statistical analyses were performed in SAS 9.4.

Cell lines and materials

Human prostate cancer cell lines were cultured as described previously (35). Briefly, LNCaP (CVCL_0395) and CWR-R1 (CVCL_4833) cells were maintained in RPMI1640 ATCC Modification (Gibco) supplemented with 10% FBS and 1% penicillin/streptomycin (P/S; Corning). LAPC4 (CVCL_4744) cells were maintained in Iscove's modified Dulbecco's medium (Hyclone) supplemented with 10% FBS, 1% P/S, and 1 nmol/L R1881 (Sigma-Aldrich). VCaP (CVCL_2235) cells were maintained in DMEM (Gibco), supplemented with 10% FBS and 1% P/S. BT474 (CVCL_0179) cells were maintained in McCoy's 5a Modified medium (Gibco) supplemented with 20% FBS and 1% P/S. MDA PCa 2b (CVCL_4748) cells were maintained in BRFF-HPC1

media (Athena ES) supplemented with 20% FBS and 1% P/S. ENZ^R cells were maintained in growth medium supplemented with 10 µmol/L enzalutamide (ENZ) as described previously (35). VCaP (CRL-2876), MDA PCa 2b (CRL-2422), and BT474 (HTB-20) cells were obtained from ATCC. LNCaP, LAPC4, and CWR-R1 cells were generously provided by Dr. John Isaacs at Johns Hopkins University (Baltimore, MD) and validated as described previously (35). Cell authentications were performed at The University of Arizona Genetics lab, and all cell lines were confirmed as negative for *Mycoplasma* using the Universal Mycoplasma Detection Kit (ATCC). For phospho-HER3 detection, cells were treated with 100 ng/mL NRG1 (Peprotech) for 15 minutes. Patritumab was obtained from MedChemExpress. For proliferation assays, cells were treated with 10 µmol/L patritumab for 5 days. For Western blot validation, cells were treated with 10 µmol/L patritumab for 2 hours with or without 100 ng/mL NRG1 for 15 minutes. Lentiviral vectors pLV[Exp]-Puro-EF1A>hERBB3 (HER3-OE, VB900139-0636jnd) and pLV[Exp]-Puro-EF1A>ORF_Non-targeting stuffer (NT, VB010000-9298rtf) were obtained from Vector Builder. Each vector was independently incubated at a multiplicity of 8 with 5 mg/mL polybrene for 20 minutes in OPTI-MEM (Thermo Fisher Scientific). Virus was added to cells for 24 hours, followed by 24 hours recovery in full media. Infected cells were selected in 1 µg/mL of puromycin (Thermo Fisher Scientific).

Western blot analysis

Western blotting was performed as described previously (35). Briefly, cells were collected in RIPA buffer, sonicated, and protein was quantified using the Pierce BCA kit (Thermo Fisher Scientific). A total of 40 µg of lysate was run on 4%–15% Mini-PROTEAN precast polyacrylamide gels (Bio-Rad). Protein was transferred to nitrocellulose membranes and blocked according to the antibody specifications (5% BSA or nonfat milk). Primary antibodies used are detailed in Supplementary Table S3. Membranes were incubated with secondary antibodies—goat anti-rabbit IRDye 800 CW (LI-COR Biosciences, catalog no. 926-32223, RRID:AB_621845) or goat anti-mouse IRDye 680 (LI-COR Biosciences, catalog no. 926-32222, RRID:AB_621844)—and images were captured using an infrared Odyssey scanner (LI-COR Biosciences).

Cell proliferation assay

CWR-R1 and LNCaP (NT, HER3-OE) cells were plated at a density of 10,000 cells per well and MDA PCa 2b cells at 15,000 cells per well in 96-well plates. Cells were treated 24 hours later, and proliferation analysis was performed using IncuCyte S3 Live-Cell Analysis System imaging every 4 hours for 68–72 hours (Sartorius). Analysis was completed in the Incucyte software with cell counts normalized to initial number of cells (hour 0). A two-way ANOVA with Tukey multiple comparison test was used for statistical analysis.

Data availability

Publicly available data were accessed through dbGaP with the following accession numbers: phs001141.v1.p1, phs000909.v1.p1, phs000915.v2.p2, phs000310.v1.p1, phs000985.v1.p1, phs000443.v1.p1, and phs001698.v1.p1 (25–30). For the UI Health cohort, the data that support the findings of this study are available on request from the corresponding author. The data are not publicly available because the subjects were not consented for the publication or distribution of their raw sequencing data.

Results

***ERBB3* OE is enriched in a diverse population of patients with CSPC**

The UI patient cohort was procured using an IRB-approved retrospective analysis of next-generation sequencing results for patients with prostate cancer. Sequencing data for RNA expression were available in 33 patients. The median age of patients in the UI cohort was 63.4 (range: 51.9–77.7 years; **Table 1**). Twenty-three of 33 (69.7%) UI patients self-reported as AA/Black and 7/33 (21.2%) self-reported as non-AA/White, with 4/7 (57.1%) reporting Hispanic ethnicity (**Table 1**). Within this population, 69.7% of patients presented with *de novo* metastases, indicating advanced disease at diagnosis (**Table 1**). Moreover, 72.3% of patients had a Gleason score sum greater than or equal to 8, while 15.2% had Gleason score sum less than or equal to 7 (12.1% unknown Gleason score; **Table 1**). Of the 33 patients, 3 were excluded from further analysis (1 due to insufficient RNA yield during tissue processing, 2 due to progression to CRPC prior to sequencing). Ultimately, all analyzed samples ($n = 30$) were sequenced from treatment-naïve tumors.

Using a positivity threshold identified during genomic analysis, *ERBB3* OE was detected in 40% of patients (**Fig. 1A**). High *ERBB3* expression was associated with more advanced disease; *ERBB3* expression was significantly higher in patients with *de novo* metastases (M1) at diagnosis compared with patients without metastases (M0; **Fig. 1B**). Applying the same threshold for *ERBB3* OE determined above, 48% of *de novo* metastatic samples had *ERBB3* OE compared with 14% OE in localized disease (**Fig. 1B**). Interestingly, there was no association between *ERBB3* expression and Gleason score, suggesting *ERBB3* as a prognostic marker, independent of Gleason score (**Fig. 1C**). Canonical alterations associated with prostate cancer (*TPMRSS2-ERG*, *PTEN*, *SPOP*, *TP53*, etc.) were assessed in the UI patient cohort, where there were no consistent alterations concurrent with *ERBB3* OE (Supplementary Table S4). One of the unique demographic characteristics of this patient population is self-reported racial and ethnic diversity. Because race is a risk factor for aggressive disease, we stratified *ERBB3* OE by race and ethnicity, resulting in 1/3 (33%) of self-identified White patients with *ERBB3* OE, and 11/27 (41%) of self-identified Black and Hispanic patients with *ERBB3* OE. This suggests a potential enrichment of *ERBB3* expression in underrepresented minority populations. Though the overall sample size is relatively small ($n = 30$), the UI cohort provides an opportunity to assess the predictive and prognostic implications of *ERBB3* OE, as well as its association with advanced prostate cancer in a racially diverse population.

To determine whether the relationship between *ERBB3* OE and race was consistent in larger datasets, we compiled prostate cancer RNA-seq data from publicly available dbGaP samples. Self-reported race was not available in these datasets, so we applied a SNP admixture analysis to determine ancestry in 664 patient samples. *ERBB3* expression was assessed in prostate cancer progression (benign $n = 61$, localized $n = 125$, metastases $n = 310$, CRPC $n = 130$) and benign prostatic hyperplasia ($n = 38$), where *ERBB3* was significantly higher in primary localized prostate cancer and metastases versus benign (**Fig. 1D**). Consistent with our finding in the UI cohort, *ERBB3* normalized expression (TPM) was significantly higher in AFR samples versus NFE in both localized and metastatic disease, suggesting this relationship is not stage specific (**Fig. 1E and F**). Pie charts depicting the percent of *ERBB3* OE in AFR versus NFE showed a higher proportion of *ERBB3* OE in AFR versus NFE in both localized prostate cancer and metastases (**Fig. 1E and F**). There was not a statistically significant difference between *ERBB3* TPM in AA samples from UI cohort versus dbGaP samples, suggesting a direct comparison is possible (Supplementary

Fig. S1A). Taken together, these data suggest that *ERBB3* RNA OE occurs in a subset of castration-sensitive prostate samples, where it was associated with more aggressive disease stage at diagnosis and was enriched in Black/AFR patients versus White/NFE. Determining the significance of *ERBB3* OE in CSPC could thus provide insight into better treatment strategies for this subset of patients with aggressive disease.

HER3 expression inversely correlates with intraprostatic DHT in Black/AA men

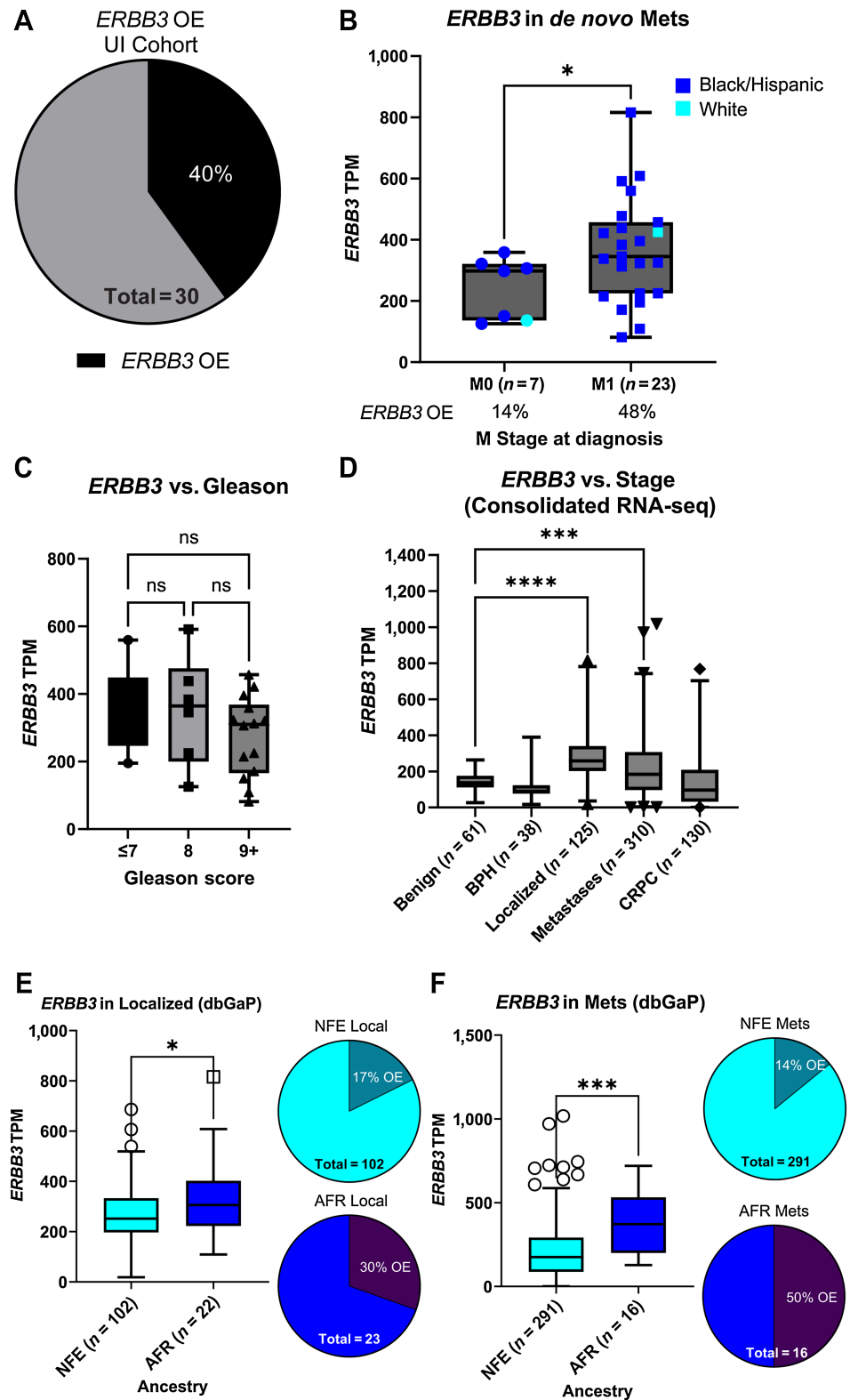
To address *ERBB3* expression at the protein level, we quantified *ERBB3* protein (HER3) in treatment-naïve primary prostate samples from a diverse patient cohort for which we also have quantification of intraprostatic androgen concentrations (ref. 34; **Table 2; Fig. 2**). HER3 positivity was quantified in luminal epithelial cells of prostate cancer cores, with normal tissue exclusion (**Fig. 2A**). HER3 percent positivity was calculated for each core, where there was no significant difference in expression between Black/AA and White patients (**Fig. 2B**). Because of the established cross-talk between HER3 and AR signaling, we assessed the relationship between intraprostatic androgen (DHT) and HER3 in these patients. Interestingly, there was a significant inverse relationship between HER3 expression and intraprostatic DHT in Black/AA patients that did not exist in White patients (**Fig. 2C and D**). This observation could be clinically significant in the context of racial disparity in prostate cancer mortality for Black men in the United States (1, 6). Because ARSIs competitively bind AR to block DHT, these therapies may not be as effective in low DHT conditions, potentially warranting alternative therapeutic strategies in these patients. A better understanding of the signaling pathways active downstream of *ERBB3*/HER3 in Black/AA compared with White men could provide insight into its role in prostate cancer etiology and disparities.

Androgen signaling is prevalent when *ERBB3* is overexpressed

To determine the signaling pathways that are associated with *ERBB3* OE, we performed GSEA on *ERBB3*-high (top quartile) versus *ERBB3*-low (bottom quartile) samples from the UI and consolidated publicly available RNA-seq samples (dbGaP; **Fig. 3**). Demographics for *ERBB3*-high and -low quartiles from the UI cohort and dbGaP samples used for GSEA show most of the UI cohort patients self-identified as Black/AA, and most of the patients accessed from dbGaP were White/NFE according to the admixture analysis (**Fig. 3A**). For the UI cohort, Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment plots identified androgen response (normalized enrichment score, NES = 1.77; FDR = 0.09) to be enriched in *ERBB3*-high samples, and inflammatory response (NES = -1.69; FDR = 0.16) to be enriched in *ERBB3*-low samples (**Fig. 3B**). Similarly, in the publicly available RNA-seq samples (dbGaP), KEGG pathway analysis showed enrichment of androgen response (Hallmark; NES = 1.79, FDR = 0.09) in *ERBB3*-high samples and enrichment of epithelial-mesenchymal transition (EMT; Hallmark; NES = -1.85, FDR = 0.06) in *ERBB3*-low samples. To further investigate the association between the androgen response pathway and *ERBB3* OE, we selected several canonical androgen response pathway genes to compare expression in *ERBB3*-high versus -low samples (**Fig. 3C**). As expected, *ERBB3*-high samples had a significantly higher expression of *ERBB3*, *KLK3*, *NKX3-1*, *TPMRSS2*, and *ZBTB10* in both the UI and dbGaP samples (**Fig. 3C**). There was not a statistically significant difference in other receptor tyrosine kinase family members (*ERBB2*, *EGFR*) or ligand NRG1 in *ERBB3*-high versus -low (Supplementary Fig. S1B). Interestingly, AR transcript was significantly higher in the *ERBB3*-high samples from dbGaP, but not in the UI cohort (**Fig. 3C**). Finally,

Figure 1.

ERBB3 OE is enriched in a diverse patient population with CSPC and is associated with advanced disease at diagnosis. **A**, ERBB3 was overexpressed in 40% of patients in the UI cohort ($n = 30$). **B**, When stratified for M stage of disease at diagnosis, ERBB3 expression was significantly higher in patients with *de novo* metastatic (M1, $n = 23$) versus M0 ($n = 7$). Using a positivity threshold of 350 TPM, 14% of M0 patients had ERBB3 OE, while 48% of patients with *de novo* metastatic disease at diagnosis had ERBB3 OE. Race/ethnicity for each patient is shown with colored dots (Black/Hispanic = purple, White = blue). **C**, When stratified for Gleason score at diagnosis, ERBB3 expression was not significantly different between Gleason score ≤ 7 , 8, or 9+ in the UI cohort. **D**, With additional samples from consolidated publicly available RNA-seq datasets, ERBB3 expression was significantly higher in localized prostate cancer ($n = 125$) and metastases ($n = 310$) versus benign ($n = 61$). **E**, In the localized prostate cancer samples from consolidated RNA-seq data, ERBB3 expression was significantly higher in samples with SNP profiles indicative of AFR ancestry ($n = 22$) versus NFE ancestry ($n = 102$). Applying the positivity threshold for ERBB3 OE to localized sample set showed 17% ERBB3 OE in NFE ancestry and 30% ERBB3 OE in AFR ancestry. **F**, In the metastatic samples from consolidated RNA-seq data, ERBB3 expression was significantly higher in samples with SNP profiles indicative of AFR ancestry ($n = 16$) versus NFE ancestry ($n = 291$). Applying the positivity threshold for ERBB3 OE to localized sample set showed 14% ERBB3 OE in NFE ancestry and 50% ERBB3 OE in AFR ancestry. TPM, transcripts per million; *, $P < 0.05$; ***, $P < 0.01$; ****, $P < 0.0001$.



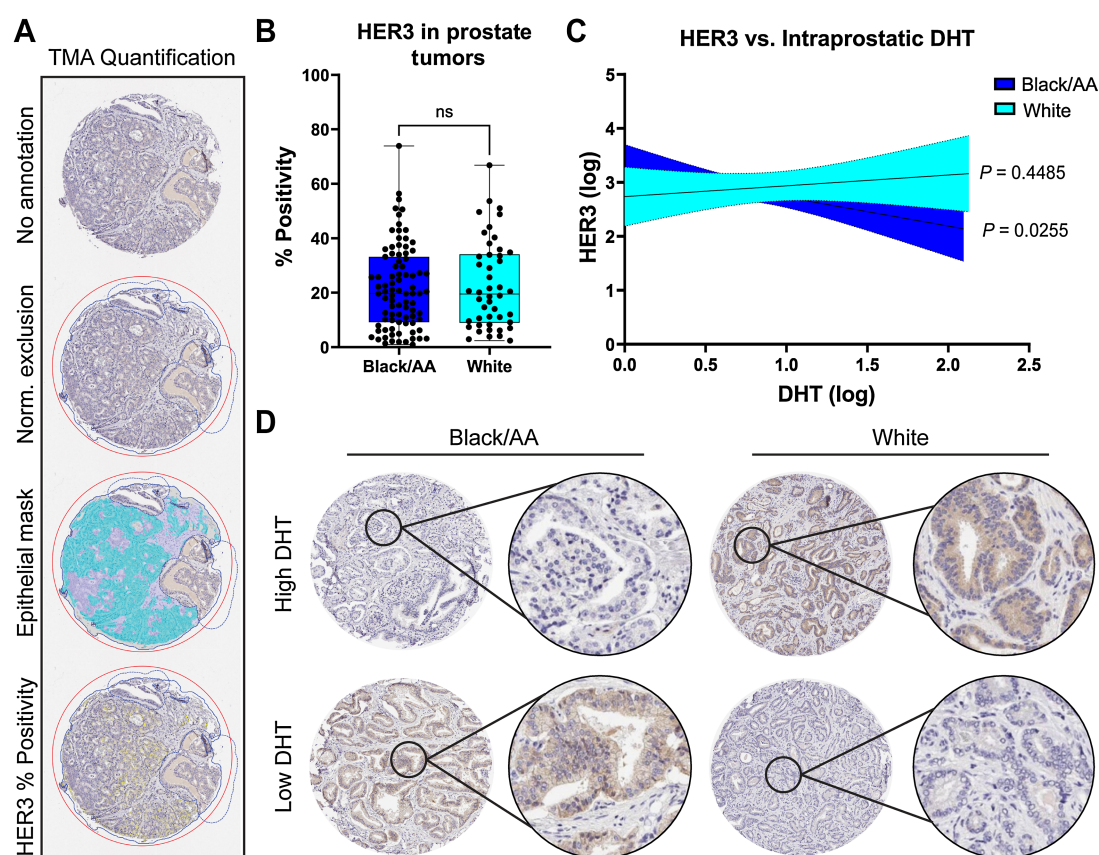


Figure 2.

HER3 expression inversely correlates with intraprostatic DHT in AA men. **A**, A TMA containing prostate tumor specimens from AA and White men was stained for HER3 and analyzed for HER3 positivity. Representative images of the TMA quantification pipeline are shown; normal areas of the cores were excluded from analysis (norm. exclusion) and HER3 was specifically quantified in epithelial cells (epithelial mask). **B**, HER3 percent positivity was not significantly different between AA and White prostate tumor cores (ns = not significant). **C**, HER3 expression (log) was negatively correlated with intraprostatic DHT (log) in AA men ($P = 0.0255$), but not in White men ($P = 0.4485$). **D**, Representative images of HER3 staining (brown) in AA and White cores with high versus low intraprostatic DHT. Hematoxylin was used as a nuclear counterstain (blue).

leading edge analysis of the genes driving the enrichment of androgen response (Hallmark) were compared between the UI cohort and samples from dbGaP using a heat map (Fig. 3D). A total of 31 leading edge genes were similar between the two datasets, seven genes were specific to the UI cohort, and 30 genes were specific to the dbGaP samples (Fig. 3E). While AR signaling was prevalent in both patient datasets, the genes driving the androgen response hallmark are different, potentially suggesting distinct pathways mediating activation of AR signaling. Given the prevalence of *ERBB3* OE, the inverse relationship between *ERBB3* and intraprostatic DHT, and the differential leading edge analysis driving androgen response hallmark in Black/AA men, it is likely that the signaling mechanisms mediating the *ERBB3*/AR axis may have differential clinical implications.

***ERBB3* expression correlates with an AR signature score and high serum PSA in Black/AA men**

To assess the clinical significance of *ERBB3*/AR signaling in Black/AA men with CSPC, we evaluated the relationship between *ERBB3*, a clinically adaptable AR signature score, and pretreatment serum PSA in the UI cohort and in publicly available dataset GSE209954. The AR signature score was based on a summation of z-scores for 34 AR-target genes conserved across reported AR chro-

matin immunoprecipitation experiments (24). A heat map of the UI cohort patients showed most *ERBB3*-high patients had a correspondingly high AR signature score (Fig. 4A). To determine whether this phenomenon was consistent in other racially diverse datasets, we compared *ERBB3* expression and AR signature score in GSE209954, which contains 33 primary prostate cancer samples (Gleason 7, 4+3) from AAM men (6). Like the UI cohort, *ERBB3* expression appeared to correlate with a positive AR signature score (Fig. 4B). To quantify this apparent relationship between *ERBB3* and AR signaling, we performed a linear regression analysis, where there was a statistically significant positive correlation between *ERBB3* expression and the summed AR signature score for the UI cohort ($R^2 = 0.3033$, $P = 0.009$) and GSE209954 ($R^2 = 0.5962$, $P < 0.0001$; Fig. 4C). We next assessed the relationship between *ERBB3* expression and pretreatment serum PSA in both the UI cohort and GSE209954. As expected, serum PSA was significantly increased in the *ERBB3*-high patients compared with *ERBB3*-low ($P = 0.0224$) in the UI cohort (Fig. 4D). Using both the AAM and NAAM samples from the GSE209954 dataset, we directly evaluated the interface between *ERBB3*, pretreatment serum PSA, and race. With all samples combined, there was no correlation between pretreatment serum PSA and *ERBB3* expression ($P = 0.3528$; Fig. 4E). However, when stratified by race, there was a statistically significant

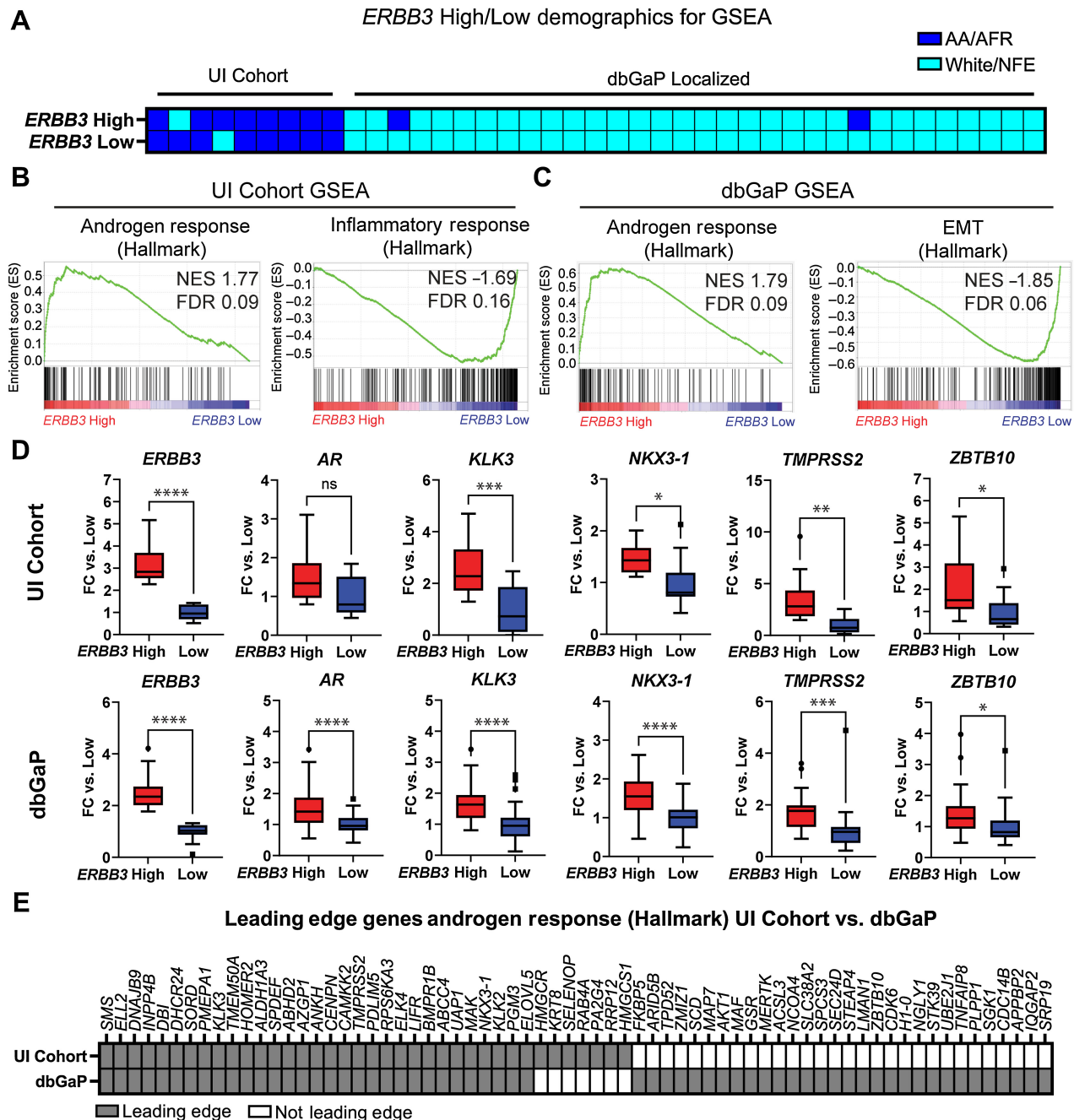
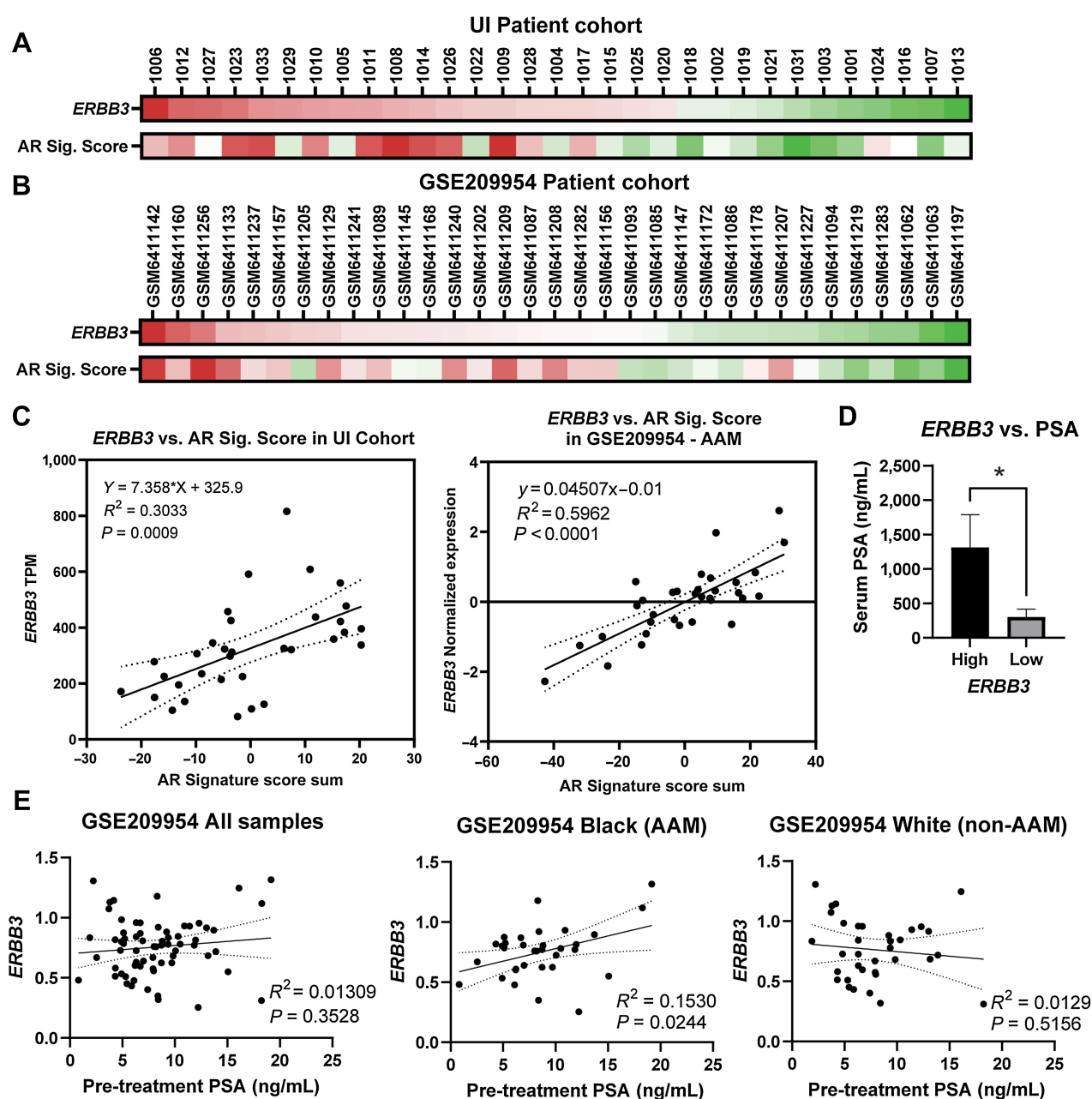


Figure 3.

ERBB3 expression is associated with androgen signaling. **A**, Demographics for *ERBB3*-high and -low quartiles from the UI cohort and consolidated publicly available RNA-seq samples (dbGaP) used for GSEA. Race for each patient analyzed is represented in a heat map where purple is AA/AFR and blue is White/NFE. **B**, GSEA between *ERBB3*-high and -low samples in the UI cohort showed enrichment of androgen response (Hallmark; NES = 1.77, FDR = 0.09) in *ERBB3*-high versus -low patients and enrichment of inflammatory response (Hallmark; NES = -1.69, FDR = 0.16) in *ERBB3*-low versus -high patients. **C**, GSEA between *ERBB3*-high and -low samples in the samples from the dbGaP cohort showed enrichment of androgen response (Hallmark; NES = 1.79, FDR = 0.09) in *ERBB3*-high versus -low patients and enrichment of EMT (Hallmark; NES = -1.85, FDR = 0.06) in *ERBB3*-low versus -high patients. **D**, To confirm the androgen response signature is prevalent in *ERBB3*-high patients, expression of select AR target genes was assessed in *ERBB3*-high versus -low patients in the UI cohort and in dbGaP samples. *ERBB3* expression was significantly higher in *ERBB3*-high patients versus *ERBB3*-low, as expected. AR transcript was not statistically different in *ERBB3*-high versus -low in the UI cohort, but significantly higher in *ERBB3*-high versus -low patients in dbGaP samples. *KLK3*, *NKX3-1*, *TMPRSS2*, and *ZBTB10* were all significantly increased in *ERBB3*-high versus -low patients. **E**, Leading edge analysis of the genes driving the enrichment of androgen response (Hallmark) were compared between the UI cohort and samples from dbGaP using a heat map where gray indicated leading edge genes and white indicated non-leading edge genes. A total of 31 leading edge genes were similar between the two datasets. Seven genes were specific to the UI cohort, and 30 genes were specific to the samples mined from dbGaP. FC, fold change; FDR, false discovery rate; NES, normalized enrichment score; ns, not significant; *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$; ****, $P < 0.0001$.

**Figure 4.**

ERBB3 expression correlates with an AR signature score and high serum PSA in AA men. **A**, *ERBB3* (log) TPM values for each patient from the UI cohort are shown in a heat map alongside the log transformed TPM values of the AR target genes that make up the AR signature score. **B**, *ERBB3* (log) TPM values for each patient from GSE209954 are shown in a heat map alongside the log transformed TPM values of the AR target genes that make up the AR signature score. **C**, A simple linear regression of *ERBB3* expression (TPM) and AR signature score sum showed a statistically significant positive correlation between *ERBB3* expression and AR signaling in the UI cohort ($R^2 = 0.3033$, $P = 0.0009$) and GSE209954 ($R^2 = 0.5962$, $P < 0.0001$). Each patient is represented by a dot with the best-fit line (solid line) bracketed by the 95% confidence intervals (dashed lines). **D**, Pretreatment serum PSA (ng/mL) was significantly higher in the *ERBB3*-high patients from the UI cohort versus *ERBB3*-low (*, $P < 0.05$). **E**, In external dataset GSE209954 containing AAM and non-AAM patients, *ERBB3* expression was not significantly correlated with pretreatment PSA ($R^2 = 0.01309$, $P = 0.3528$) in all samples. However, when stratified by race, there was a significant positive correlation between *ERBB3* expression and pretreatment PSA in AAM patients ($R^2 = 0.1530$, $P = 0.0244$) that did not exist in non-AAM patients ($R^2 = 0.0129$, $P = 0.5156$). Each patient is represented by a dot with the best-fit line (solid line) bracketed by the 95% confidence intervals (dashed lines).

correlation between *ERBB3* expression and serum PSA in Black/AAM men ($P = 0.0244$) that did not exist in the NAAM samples ($P = 0.5156$; **Fig. 4E**). Because all patients analyzed from this dataset have the same Gleason score (4+3) and disease stage [$<T2c$, (6)] this

relationship between PSA and *ERBB3* appears to be independent of disease burden. In addition, the UI cohort and external dataset GSE209954 both suggest that pretreatment serum PSA is significantly higher in Black/AA patients with high *ERBB3*, but not White patients.

These data support our earlier finding that AR signaling is increased in Black/AA patients with high *ERBB3* expression, and further support the idea that the cross-talk between *ERBB3* and AR is clinically important as a prognostic biomarker in diverse patients with prostate cancer.

HER3 promotes ENZ resistance in prostate cancer models

Our data demonstrate that *ERBB3*/HER3 is associated with androgen signaling in a subset of patients with CSPC, suggesting it may play a role in sensitivity to AR-targeted therapeutics such as ENZ. To directly assess the functional relationship between HER3 and ENZ sensitivity, we performed mechanistic gain- and loss-of-function experiments in prostate cancer cell line models. In a panel of prostate cancer cell lines, HER3 was highly expressed in CWR-R1 and MDA PCa 2b cells and lowly expressed in LNCaP and VCaP cells (Fig. 5A). In previously published ENZ-adapted (ENZ-resistant, ENZ^R) CWR-R1, LNCaP, and VCaP cells treated with HER3 ligand NRG1, phospho-HER3 (pHER3) was increased in ENZ^R lines versus parental, suggesting increased HER3 responsiveness in models of ENZ resistance (Fig. 5B). phospho-HER2 (pHER2)/HER2 were also increased in CWR-R1 ENZ^R versus parental, but not LNCaP or VCaP ENZ^R (Fig. 5B). Conversely, in androgen-adapted LNCaPs grown in R1881 for 6 months, pHER3/HER3, but not pHER2/HER2, were decreased in the high androgen condition (10 nmol/L R1881) compared with control (0 nmol/L; Fig. 5C). To determine whether HER3 is necessary for ENZ resistance, we assessed the growth of CWR-R1 and MDA PCa 2b cells after treatment with HER3 inhibitor patritumab in combination with ENZ (Fig. 5D; Supplementary Fig. S2A). ENZ alone did not decrease proliferation, but when cotreated with patritumab, cell proliferation decreased significantly compared with either treatment alone ($P < 0.0001$; Fig. 5D). Western blots confirmed reduction of pHER3 with patritumab (Fig. 5E; Supplementary Fig. S2B). Interestingly, proliferation of CWR-R1 and MDA PCa 2b cells was decreased after cotreatment with ENZ and patritumab–deruxtecan, an antibody–drug conjugate (ADC) that is under clinical development in other disease states (Supplementary Fig. S2C and S2D). To determine whether HER3 is sufficient for ENZ resistance, we ectopically overexpressed HER3 in LNCaP cells and assessed proliferation after treatment with ENZ (Fig. 5F; Supplementary Fig. S2). In control cells (non-targeted vector, NT), cell proliferation decreased after treatment with 10 μ mol/L ENZ; however, proliferation in the HER3-OE cells did not decrease after treatment with ENZ, suggesting HER3-OE is sufficient for ENZ resistance (Fig. 5F). Western blots confirmed OE of HER3 compared with the non-targeted control vector (Fig. 5G; Supplementary Fig. S2). Taken together, these results indicate that HER3 is necessary and sufficient for ENZ resistance and support a clinical approach targeting HER3 to prevent or reverse ENZ resistance.

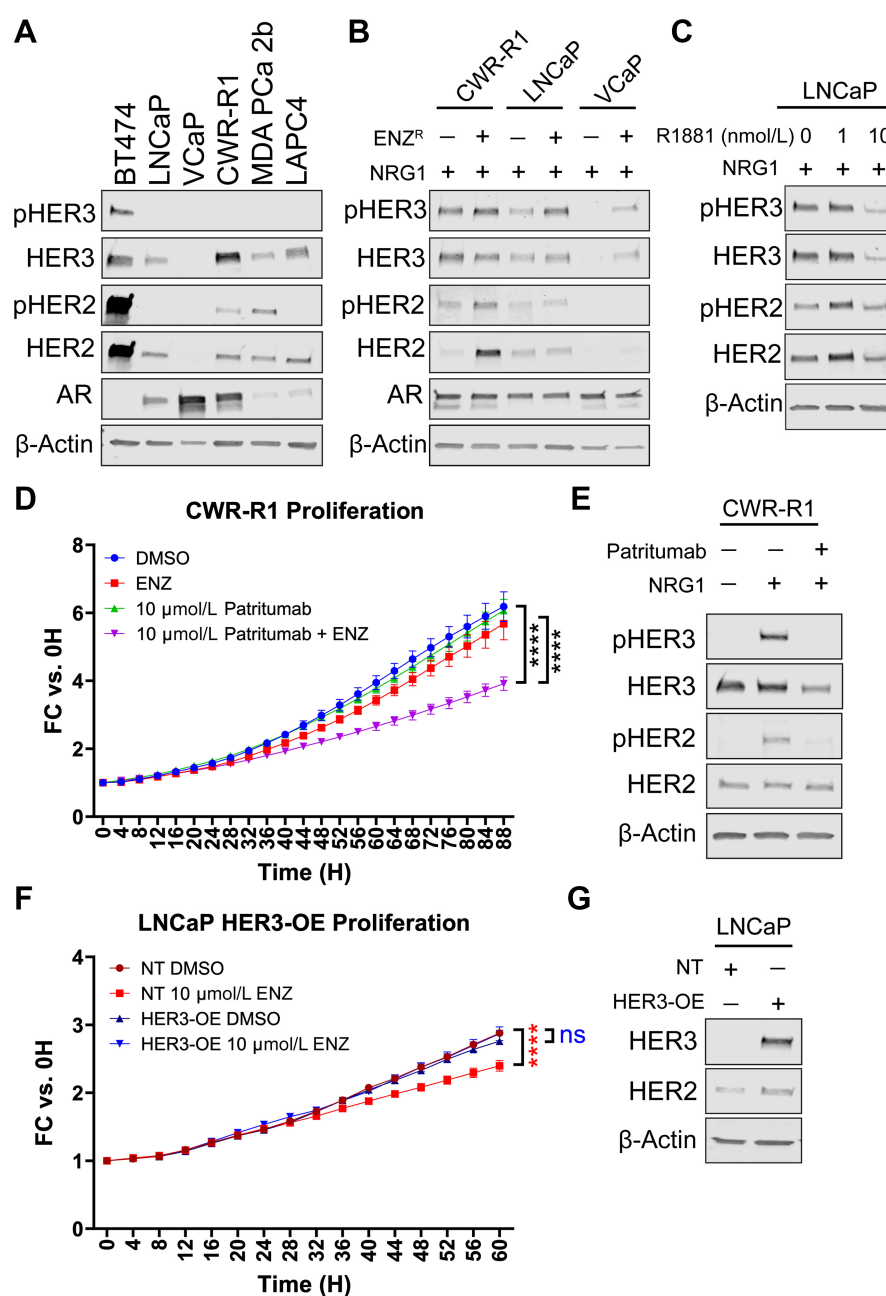
Discussion

In this study, we identified *ERBB3* OE in a subset of CSPC diagnoses. This mRNA OE was enriched in Black/AA men and was associated with more advanced disease at clinical presentation. Bioinformatic analysis of this *ERBB3*-high population revealed a relationship between *ERBB3* and increased AR signaling. This association between *ERBB3* and the androgen signaling axis was confirmed in external datasets and in patient tissue samples where HER3 expression was negatively correlated with intraprostatic DHT levels in AA patients.

Taken together, these data suggest that Black/AA patients with *ERBB3* OE have significantly higher pretreatment PSA and AR signaling, but low intraprostatic DHT. Mechanistically, this inverse relationship could suggest a pre-therapeutic ligand-independent activation of AR signaling. In this context, ARSIs, which directly compete with DHT for binding to AR, may not be as effective or durable, implicating high *ERBB3*/HER3 expression as a biomarker for patients who will benefit from intensification of therapy in the castration-sensitive setting. Here, we demonstrated that HER3 was necessary and sufficient to confer ENZ resistance in prostate cancer cell line models and that these models accurately represent the inverse relationship we observed between androgens and HER3 from patient tissue analyses. A better understanding of *ERBB3* OE and its role as a potential mechanism of AR-targeted therapeutic resistance could improve therapeutic outcomes in this subset of patients.

Receptor tyrosine kinases have been implicated in AR signaling as a mechanism for hormone-independent prostate cancer for decades (36). However, phase II clinical trials targeting EGFR (ERBB1) and HER2 (ERBB2) in CRPC have not yet proven to be clinically significant (37–41). Recently, HER3 (*ERBB3*) has been implicated in CRPC, where it was associated with disease recurrence and shorter overall survival (9). Following promising preclinical data, several phase I clinical trials targeting HER3 using mAbs or ADCs are currently recruiting patients with CRPC (NCT05057013, NCT05588609, NCT05785741). Our study suggests an additional setting for targeting HER3: CSPC. However, *ERBB3*/HER3 data in primary prostate cancer have historically been inconsistent. In some studies, *ERBB3*/HER3 increases throughout prostate cancer progression where it is associated with poor prognosis, while others show *ERBB3*-low primary prostate cancer radical prostatectomy samples were associated with worse prognosis (9, 42, 43). In addition, several studies implicate nuclear HER3 as either an indicator of progression to hormone-refractory prostate cancer, or the lack of nuclear HER3 as predictor of biochemical recurrence (44, 45). Potential explanations for the controversy surrounding *ERBB3*/HER3 in prostate cancer is that it may serve as a biomarker in a specific cellular localization (nuclear vs. cytosolic vs. membrane) or may only be relevant in a particular subset of patients. Further investigation into stratification for *ERBB3*/HER3 biomarker and/or drug targeting studies should be considered moving forward.

In the UI patient cohort, we see *ERBB3* OE as a subtype of CSPC that has not been described previously. Because our patient population is more racially diverse than many previous prostate cancer sequencing cohorts, it is possible that *ERBB3* OE has been overlooked in previous CSPC studies due to lack of diversity in samples analyzed. Recently, an increasing number of studies are investigating racial disparities in prostate cancer incidence and mortality (1, 6, 46). The cause of the racial disparity in prostate cancer is multifactorial and not yet fully elucidated, stemming from a combination of inequitable access to health resources, lack of representation in research, and epigenetic sequelae of environmental factors (e.g., stress, metabolism, diet). Although race is an inherently social category, biogeographical ancestry analyses have shown that there is a strong concordance between molecular-based ancestry groups and racial groups (47, 48). Identification of molecular expression and activity patterns that vary by racial group may provide a key to addressing outcomes related to racial disparities, as currently existing treatments have not been developed with attention to the inclusion of diverse patient populations. In one recent study, racial disparities were assessed in veterans who have relatively equal access to health care through Veterans Affairs; despite this equal access setting with standardized treatment regimens,

**Figure 5.**

HER3 promotes ENZ resistance in prostate cancer models. **A**, Protein expression of phospho-HER3 (pHER3), HER3, phospho-HER2 (pHER2), HER2, AR, and beta-actin was assessed in an array of prostate cancer cell lines (LNCaP, VCaP, CWR-R1, MDA PCa 2b, LAPC4). BT474 (breast cancer) was used as a positive control for HER2/3 expression and beta-actin was used as a loading control. **B**, Protein expression of pHER3, HER3, pHER2, HER2, AR) was assessed in ENZ-resistant (ENZ^R) CWR-R1, LNCaP, and VCaP cells. ENZ^R cells were maintained in 0 μmol/L ENZ and all cells were treated with 100 ng/mL NRG1 for 15 minutes prior to protein collection. Beta-actin was used as a loading control. **C**, LNCaP were maintained in 0, 1, or 10 nmol/L R1881 for 6 months to create an androgen-adapted cell line. Cells were treated with 100 ng/mL NRG1 for 15 minutes prior to protein collection. pHER3 and HER3 were decreased in the high androgen condition (10 nmol/L R1881) compared with control (0 nmol/L R1881). pHER2 and HER2 were increased in low androgen (1 nmol/L R1881), but not different in the 10 nmol/L condition versus control. Beta-actin was used as a loading control. **D**, Cell proliferation was assessed in CWR-R1 cells after treatment with HER3 inhibitor patritumab in combination with ENZ. ENZ alone did not decrease proliferation, but when co-treated with patritumab, cell proliferation decreased significantly compared with either treatment alone ($n = 5$). **E**, Western blots confirmed reduction of pHER3 with patritumab (10 μmol/L for 2 hours) after NRG1 stimulation (100 ng/mL for 15 minutes). pHER2 was also decreased after patritumab treatment, but to a lesser degree. Beta-actin was used as a loading control. **F**, A representative cell proliferation curve in LNCaP HER3 overexpression (HER3-OE) versus non-targeted control (NT) vectors after treatment with 10 μmol/L ENZ. Proliferation decreased in the NT control cells after treatment with 10 μmol/L ENZ; however, the HER3-OE cells did not decrease after treatment with ENZ, suggesting HER3-OE is sufficient for ENZ resistance ($n = 3$). **G**, Western blots confirmed OE of HER3 compared with NT control. A two-way ANOVA with Tukey multiple comparison test was used for statistics; ****, $P < 0.0001$; ns, not significant.

Black/AA men were younger at diagnosis and had higher AR signaling, more advanced disease, and increased residual metastatic burden compared with White men (46, 49). However, a general lack of inclusion of diverse patients in biomarker identification studies has resulted in information disparity for prognosis and targeted therapeutic options. Many of the parameters assessed in the current study support previous findings for Black/AA men with prostate cancer including more advanced disease at diagnosis and increased AR signaling. In this context, *ERBB3* OE may serve as a biomarker for aggressive disease, stratifying patients that will most likely benefit from intensification of therapy. Validating *ERBB3* OE as a biomarker and identifying additional genomic alterations within this vulnerable population has high potential to impact patient care and improve treatment outcomes.

While our data generally support other recent findings, there are some limitations. First, clinical outcomes for the UI cohort are not yet available. While initial analyses of this cohort and others suggest a relationship between *ERBB3* OE and overall survival (Supplementary Fig. S3A) and time to progression (Supplementary Fig. S3B), investigation in datasets with more detailed outcomes is warranted to develop *ERBB3* OE as a biomarker and/or target in CSPC. Second, retrospective disparities research is impeded by a lack of patient demographic information in comprehensive datasets like dbGaP. To address this, we utilized an admixture analysis using SNP data to determine ancestry. While this is a useful bioinformatics tool, it is limited by the lack of information regarding other factors contributing to disparities (e.g., access to health care, environmental factors). Therefore, it should only be used as a tool to assess genetic drivers of disparity, and other sociologic studies should address additional contributing factors. Third, based on our data, *ERBB3* mRNA expression could be a biomarker for poor prognosis; however, this differential expression was not captured at the protein level by IHC analysis. This inconsistency could be due to technical limitations of IHC, where the total protein expression may not be the biomarker, but rather functionally active membrane-bound HER3 would correlate better with the mRNA data for Black/AA men. In addition, the TPM values for *ERBB3* showed expression in all samples, with higher expression in the *ERBB3* OE subset. It is possible that IHC is not reflecting these quantitative changes due to lack of assay sensitivity. Finally, we demonstrated that *ERBB3* OE was associated with increased AR signaling despite lower intraprostatic DHT in Black/AA men. This is somewhat anti-intuitive in terms of mechanisms for CRPC progression; AR signaling is frequently increased in CRPC but has been shown to coincide with increased intraprostatic DHT (50, 51). In our *ERBB3* OE samples, the increased AR signaling does not appear to be mediated by the ligand, DHT, which may indicate an alternative activation of AR signaling in Black/AA men. This ligand-independent activation of AR by receptor tyrosine kinases has been proposed previously in prostate cancer cell line models, where HER2/HER3 dimerization induced AR activation in an androgen-depleted environment (12); however, the clinical implications of this aberrant AR signaling in the context of racial disparities has not been explored. This prospective model, and the functionality and therapeutic implications of *ERBB3*/HER3 in racially diverse patients with CSPC, should be further investigated.

Here, we have provided evidence of *ERBB3* OE in a subset of diverse patients with CSPC where it correlated with the AR signaling pathway activity. Additional bioinformatic and immunohistologic analyses showed that this AR signaling activity occurs despite stable expression of the androgen receptor, and at the protein level, in low intraprostatic

androgen conditions. Taken together, these data suggest that this AR signaling is being maintained despite low availability of androgen. In terms of clinical management, this unique induction of AR signaling may indicate that AR-targeted therapies alone are not sufficient for this subtype of CSPC and that *ERBB3* OE may be a biomarker for patients that will benefit from intensification of therapy in the castration-sensitive setting. Furthermore, our preclinical data suggest that directly targeting HER3 in prostate cancer with *ERBB3*/HER3-OE could increase sensitivity to ENZ. On the basis of these findings, prospective validation studies for *ERBB3* OE in CSPC clinical trials and/or clinical trials targeting HER3 in conjunction with AR-targeted therapies have strong potential to better clinical management of prostate cancer. Importantly, *ERBB3* OE is enriched in a diverse patient cohort, potentially suggesting a contribution to racial disparities, yet also informing potential targeted therapeutic strategies in diverse individuals. With better inclusion of more diverse patient cohorts in future analyses, we can close this knowledge gap, improving stratification and selection of patients with prostate cancer who may benefit from intensification and personalization of therapy.

Authors' Disclosures

R.H. Nguyen reports personal fees from Genomeweb and Merck outside the submitted work. N.M. Reizine reports grants from ACS during the conduct of the study, as well as personal fees from Sanofi, Exelixis, Janssen, AstraZeneca, EMD Serono, Merck, and Tempus outside the submitted work. No disclosures were reported by the other authors.

Authors' Contributions

J.E. Vellky: Conceptualization, data curation, formal analysis, validation, investigation, visualization, methodology, writing—original draft, project administration, writing—review and editing. **B.J. Kirkpatrick:** Conceptualization, data curation, validation, writing—original draft, writing—review and editing. **L.C. Gutgesell:** Data curation, software, formal analysis, validation, investigation, methodology. **M. Morales:** Conceptualization, data curation, formal analysis, methodology, writing—review and editing. **R.M. Brown:** Data curation, formal analysis, visualization, methodology. **Y. Wu:** Data curation, software, formal analysis. **M. Maienschein-Cline:** Resources, data curation, formal analysis, methodology. **L.D. Notardonato:** Data curation, software, methodology. **M.S. Weinfeld:** Resources, software, supervision, methodology. **R.H. Nguyen:** Conceptualization, resources, data curation, software, methodology, writing—original draft, writing—review and editing. **E. Brister:** Data curation, formal analysis, validation, investigation, visualization, methodology. **M. Sverdllov:** Resources, data curation, formal analysis, methodology. **L. Liu:** Data curation, formal analysis, visualization, methodology. **Z. Xu:** Data curation, formal analysis, visualization, methodology. **S. Kregel:** Conceptualization, writing—original draft, writing—review and editing. **L. Nonn:** Conceptualization, resources, formal analysis, supervision, funding acquisition, investigation, methodology, project administration, writing—review and editing. **D.J. Vander Griend:** Conceptualization, resources, supervision, methodology, project administration, writing—review and editing. **N.M. Reizine:** Conceptualization, resources, data curation, software, formal analysis, supervision, funding acquisition, writing—original draft, writing—review and editing.

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Note

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